

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0868

North Augusta 138 kV
February 23, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0868. The Interconnection Customer proposes a 77.1 MW solar photovoltaic facility interconnecting at the North Augusta 138 kV substation in the Kentucky area. The request was received on September 29, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,114,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Snohomish-Manwaring 138.0 kV
Contingency: Berry Street-Wissota 138.0 kV
Loading: 139.6% (Rating: 259 MVA)
DFAX: 5.5%
Pre-project loading: 138.0%
- Facility: Wagner Lake-New Hollandale 230.0 kV
Contingency: Bay Pt-Oakland Ave 230.0 kV
Loading: 123.7% (Rating: 469 MVA)
DFAX: 9.9%
Pre-project loading: 122.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Snohomish-Manwaring 138.0 kV	Rebuild line with higher-capacity conductor	\$4,562,000	3% (\$143,000)
Wagner Lake-New Hollandale 230.0 kV	Rebuild line with higher-capacity conductor	\$14,006,000	7% (\$971,000)
	Total allocated to project		\$1,114,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.